INTEGRALE IMPROPRII. PROBLEME PROPUSE

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Folosind posibilitatea de derivare sub semnul integral, să se calculeze:

1. $F(y)=\int_0^{\frac{\pi}{2}}\frac{1}{\operatorname{tg}x}\operatorname{arctg}(y\operatorname{tg}x)dx, y\geq 0$ **R:** $F(y)=\frac{\pi}{2}\ln(y+1)$
2. $I(a)=\int_0^{\frac{\pi}{2}}\ln\frac{1+a\sin x}{1-a\sin x}\cdot\frac{dx}{\sin x}, |a|<1$ **R:** $I(a)=\pi\arcsin a$
3. $I(a)=\int_0^1\frac{\ln(1-a^2x^2)}{x^2\sqrt{1-x^2}}dx, a^2<1$ **R:** $I(a)=\pi(\sqrt{1-a^2}-1)$
4. $I(m)=\int_0^{\frac{\pi}{2}}\ln(\cos^2x+m^2\sin^2x)dx, m\in\mathbb{R}_+$ **R:** $I(m)=\pi\ln\frac{m+1}{2}$
5. $I(a)=\int_0^1\frac{\ln(1-ax^2)}{\sqrt{1-x^2}}dx, |a|<1$ **R:** $I(a)=\pi\ln(1+\sqrt{1-a})-\pi\ln 2$
6. Fie $F(x)=\int_0^x\frac{\ln(1+tx)}{1+t^2}dt,\ \ x\geq 0$. a) Calculaţi $F'(x)$; b) Calculaţi $F(x)$; c) Calculaţi $I=\int_0^{\frac{\pi}{4}}\ln(1+\operatorname{tg}x)dx$.
   R: b) $F(x)=\frac{1}{2}\ln(1+x^2)\cdot\operatorname{arctg}x$; c) $I=\frac{\pi}{8}\ln 2$.

Să se calculeze următoarele integrale cu ajutorul funcţiilor Γ şi B:

7. $\int_0^\infty \mathrm{e}^{-x^p}dx, p>0$ **R:** $\Gamma\left(1+\frac{1}{p}\right)=\frac{1}{p}\Gamma\left(\frac{1}{p}\right)$
8. $\int_0^1\sqrt{x-x^2}dx$ **R:** $\frac{\pi}{8}$
9. $\int_0^\infty\frac{dx}{1+x^4}$ **R:** $\frac{\pi\sqrt{2}}{8}$
10. $\int_0^\infty\frac{dx}{\sqrt[3]{x}(1+x)}$ **R:** $\frac{2\pi}{\sqrt{3}}$
11. $\int_0^\infty\frac{x^{m-1}}{1+x^n}dx, m,n>0$ **R:** $\frac{1}{n}\cdot\frac{\pi}{\sin\frac{m\pi}{n}}, 0<m<n$
12. $\int_0^\infty\frac{\sqrt[3]{x}}{(1+x^2)^2}dx$ **R:** $\frac{\pi}{3\sqrt{3}}$
13. $\int_0^\infty\frac{\sqrt[3]{x^5}}{(1+x^2)^4}dx$
14. $\int_0^\infty\frac{\sqrt[5]{x}}{(1+x^2)^3}dx$
15. $\int_{-\infty}^\infty \mathrm{e}^{-2x^2+x+1}dx$ **R:** $\frac{1}{2}\cdot\mathrm{e}^{\frac{9}{8}}\cdot\sqrt{2\pi}$
16. $\int_{-1}^2\frac{dx}{\sqrt[6]{(x+1)^5(2-x)}}$ **R:** 2π
17. $\int_0^1\sqrt{\ln\frac{1}{x}}dx$ **R:** $\frac{\sqrt{\pi}}{2}$
18. $I=\int_0^\infty\frac{\sqrt[4]{x}}{(x+1)^2}$ **R:** $\frac{\pi\sqrt{2}}{4}$

19. $\int\limits_0^1 x^{p+1}(1-x^m)^{q-1}\,dx, p>0, q>0, m>0$

20. $\int_1^{\mathrm{e}}(\ln x)^{m+a}\left(\ln\frac{\mathrm{e}}{x}\right)^{n-a}\cdot\frac{1}{x}dx, m,n\in\mathbb{N}, a\in(0,1)$

21. $\int_0^{\frac{\pi}{2}}\sqrt[3]{\operatorname{tg} x}dx$

22. $\int_0^{\infty} x^4(1+\frac{x^2}{2})^{-\frac{7}{2}}dx$

23. Calculaţi $\int_0^1\frac{1-x^2}{\sqrt[3]{1-x^3}}dx$.

24. Calculaţi $I_n=\int_0^1\frac{x^{2n+1}}{\sqrt{1-x^2}}dx$.

    b) Este seria $\sum\limits_{n=1}^{\infty}I_n$ convergentă ? Justificaţi răspunsul.

    R: a) $I_n=\frac{(2n)!!}{(2n+1)!!}$